

SRI VASAVI ENGINEERING COLLEGE (Autonomous)

(Permanent Affiliation to JNTUK, Kakinada),
PEDATADEPALLI, TADEPALLIGUDEM-534 101

A.Y: 2024-25

VII SEM CSE Handbook (V20 Regulation)

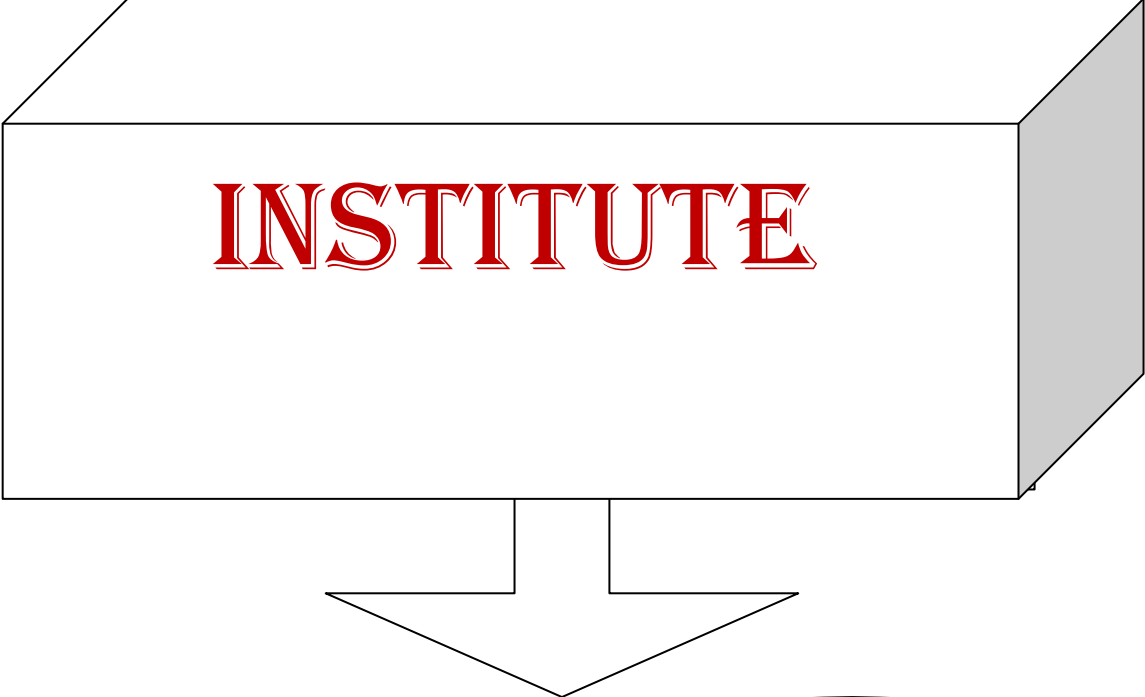


**Department of Computer Science and Engineering (Accredited by
NBA)**

Pedatadepalli, Tadepalligudem-534101, A.P

INDEX

S.NO	CONTENTS	PAGE NO.
1.	Institute Vision & Mission	4
2.	Department Vision & Mission	6
3.	Program Educational Objectives, Program Outcomes & Program Specific Outcomes	8
4.	Academic Calendar	10
5.	Class Time Table	12
6.	Course Structure	13
Lesson Plans		15 -43
7.	Elective – III: Deep Learning	15
8.	Elective – IV: NoSQL Databases	18
9.	Elective – IV: Cloud Computing	22
10.	Elective-V: Software Project Management	26
11.	Elective-V: Social Networks and Semantic Web	30
12.	Management Science	34
13.	Job Oriented Elective –III: Full Stack Technologies	37
14.	Job Oriented Elective –IV: DevOps	40



INSTITUTE



VISION MISSION

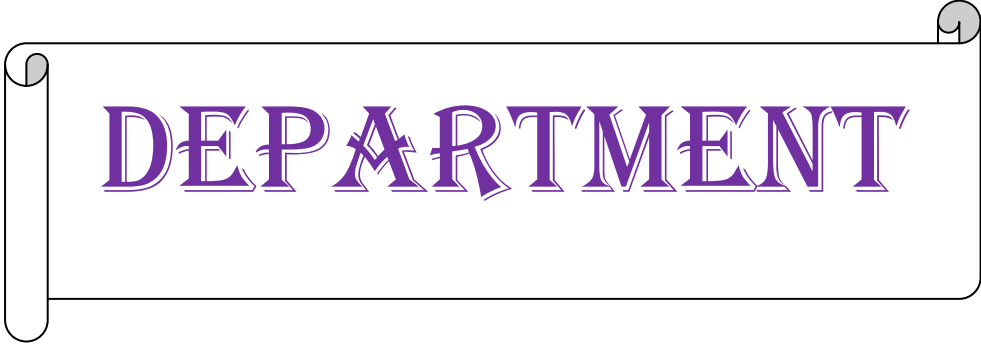
INSTITUTE VISION AND MISSION

VISION

To be a premier technological institute striving for excellence with global perspective and commitment to the nation.

MISSION

- To produce engineering graduates of professional quality and global perspective through Learner Centric Education.
- To establish linkages with government, industry and research laboratories to promote R&D activities and to disseminate innovations.
- To create an eco-system in the institute that leads to holistic development and ability for life-long learning..



DEPARTMENT



VISION MISSION

Vision:

- To evolve as a centre of academic and research excellence in the area of Computer Science and Engineering.

Mission :

- To utilize innovative learning methods for academic improvement.
- To encourage higher studies and research to meet the futuristic requirements of Computer Science and Engineering.
- To inculcate Ethics and Human values for developing students with good character

**PROGRAM
EDUCATIONAL
OBJECTIVES,
PROGRAM OUTCOMES
& PROGRAM
SPECIFIC
OUTCOMES**

Program Educational Objectives (PEOs): Graduates of this programme will

PEO 1: Adapt to evolving technology.

PEO 2: Provide optimal solutions to real time problems.

PEO 3: Demonstrate his/her abilities to support service activities with due consideration for Professional and Ethical Values.

Programme Specific Outcomes (PSOs): A graduate of the Computer Science and Engineering Program will be able to

PSO 1: Use Mathematical Abstractions and Algorithmic Design along with Open Source Programming tools to solve complexities involved in Programming. [K3]

PSO 2: Use Professional engineering practices and strategies for development and maintenance of software. [K3]

Program Outcomes (POs):

Computer Science Engineering Graduates will be able to:

1. **Engineering knowledge:** Apply the knowledge of Mathematics, Science, Engineering Fundamentals and Concepts of Computer Science Engineering to the solution of complex Engineering problems. [K3]
2. **Problem analysis:** Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of Mathematics, Natural Sciences and Computer Science. [K4]
3. **Design/development of solutions:** Design solutions for complex engineering problems and design system components or processes that meet the specific needs with appropriate consideration for the public health and safety, and the cultural, societal and environmental considerations. [K5]
4. **Conduct investigations of complex problems:** Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions. [K5]
5. **Modern tool usage:** Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex Engineering activities with an understanding of the limitations. [K3]
6. **The Engineer and society:** Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional Engineering practice. [K3]
7. **Environment and sustainability:** Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development. [K3]
8. **Ethics:** Apply ethical principles and commit to professional ethics and responsibilities and norms of the Engineering practice. [K3]
9. **Individual and team work:** Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings. [K6]
10. **Communication:** Communicate effectively on complex Engineering activities with the Engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions. [K2]
11. **Project management and finance:** Demonstrate knowledge and understanding of the Engineering and Management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments. [K6]
12. **Life-long learning:** Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change. [K1]

ACADEMIC CALENDAR

✉ : principal@srivasaviengg.ac.in
svec.a8@gmail.com



☎ : 08818- 284344, 355

SRI VASAVI ENGINEERING COLLEGE (AUTONOMOUS)

(Sponsored by Sri Vasavi Educational Society)

(Approved by AICTE, New Delhi & Permanently affiliated to JNTUK, Kakinada)

(Accredited by NAAC with 'A' Grade, Recognized by UGC under section 2(f) & 12(B))

(NBA Accreditation to B.Tech., EEE, CSE, ME and ECE Branches for 3 Years)

Pedatadepalli, **TADEPALLIGUDEM** – 534 101. W.G.Dist. (A.P)

Principal's Office
Date: 06-07-2024

Revised Academic Calendar

For IV B.Tech (VII and VIII Semesters), Academic Year 2024-25

VII Semester			
Description	From	To	Weeks
Commencement of Class Work	22.07.2024		
I Unit of Instructions	22.07.2024	14.09.2024	8 W
I Mid Examinations	16.09.2024	21.09.2024	1 W
II Unit of Instructions	23.09.2024	16.11.2024	8 W
II Mid Examinations	18.11.2024	23.11.2024	1 W
Preparation and Practicals	25.11.2024	30.11.2024	1 W
End Examinations	02.12.2024	14.12.2024	2 W
VIII Semester			
Commencement of Internship cum Project Work	16.12.2024		
Commencement of Project Work	16.12.2024	04.04.2025	16W
Thesis Submission & Seminar	07.04.2025	11.04.2025	1 W
End Viva-voce Examinations	15.04.2025	19.04.2025	1 W


PRINCIPAL

Copy to : ALL

Vision

To be a premier technological institute striving for excellence with global perspective and commitment to the nation.

Mission

- To produce Engineering graduates of professional quality and global perspective through learner-centric education.
- To establish linkages with government, industry and Research laboratories to promote R&D activities and to disseminate innovations.
- To create an eco-system in the institute that leads to holistic development and ability for life-long learning.



SRI VASAVI ENGINEERING COLLEGE (Autonomous)

Pedatadepalli, TADEPALLIGUDEM-534 101, W.G. Dist.

Department Of Computer Science & Engineering (Accredited by NBA)



CLASS CONSOLIDATED TIME TABLE

(Academic Year 2024-25)

w.e.f. 22.07-2024

Class: VII SEM

Section – A

Class Coordinator: Mr. N.V. Ratnakishor Gade

Room: G-301

Periods	1	2	3	4	1:00PM 2:00PM	5	6	7
Time Day	(09.30 AM- 10.30 AM)	(10.30 AM- 11.20 AM)	(11.20 AM- 12.10 PM)	(12.10 PM- 01.00 PM)		(02.00 PM- 02.50 PM)	(02.50 PM- 03.40 PM)	(03.40 PM- 04.30 PM)
Mon	Devops	Java Full Stack Technologies Lab			LUNCH BREAK	NOSQL	DL	MS
Tue	DL	MS	Devops	NOSQL		DL	SNSW	Devops
Wed	SNSW	SNSW	NOSQL	Devops		Java Full Stack Technologies Lab		
Thu	NOSQL	Devops	DL	NOSQL		MS	SNSW	Devops
Fri	DL	SNSW	SNSW	MS		MS	NOSQL	DL

Class: VII SEM

Section – B

Class Coordinator: Mrs. D. S L Manikanteswari

Room: G-302

Periods	1	2	3	4	1:00PM 2:00PM	5	6	7
Time Day	(09.30 AM- 10.30 AM)	(10.30 AM- 11.20 AM)	(11.20 AM- 12.10 PM)	(12.10 PM- 01.00 PM)		(02.00 PM- 02.50 PM)	(02.50 PM- 03.40 PM)	(03.40 PM- 04.30 PM)
Mon	SNSW	Java Full Stack Technologies Lab			LUNCH BREAK	CC	SNSW	Devops
Tue	Devops	CC	SNSW	DL		Devops	MS	CC
Wed	CC	DL	MS	DL		Java Full Stack Technologies Lab		
Thu	DL	SNSW	SNSW	Devops		MS	CC	DL
Fri	MS	CC	Devops	DL		Devops	MS	SNSW

Class: VII SEM

Section – C

Class Coordinator: Mr. M.V.V. Krishna

Room: G-401

Periods	1	2	3	4	1:00PM 2:00PM	5	6	7
Time Day	(09.30 AM- 10.30 AM)	(10.30 AM- 11.20 AM)	(11.20 AM- 12.10 PM)	(12.10 PM- 01.00 PM)		(02.00 PM- 02.50 PM)	(02.50 PM- 03.40 PM)	(03.40 PM- 04.30 PM)
Mon	CC	Devops	SNSW	Devops	LUNCH BREAK	DL	MS	CC
Tue	DL	Java Full Stack Technologies Lab				SNSW	SNSW	DL
Wed	SNSW	MS	DL	Devops		CC	MS	Devops
Thu	CC	DL	CC	SNSW		Devops	MS	SNSW
Fri	MS	Devops	DL	CC		Java Full Stack Technologies Lab		

Class: VII SEM

Section – D

Class Coordinator: Dr. G Tej Varma

Room: G-402

Periods	1	2	3	4	1:00PM 2:00PM LUNCH BREAK	5	6	7
Time Day	(09.30 AM- 10.30 AM)	(10.30 AM- 11.20 AM)	(11.20 AM- 12.10 PM)	(12.10 PM- 01.00 PM)		(02.00 PM- 02.50 PM)	(02.50 PM- 03.40 PM)	(03.40 PM- 04.30 PM)
Mon	MS	SPM	NOSQL	DL		Devops	SPM	DL
Tue	MS	Java Full Stack Technologies Lab				DL	NOSQL	Devops
Wed	NOSQL	Devops	SPM	DL		Devops	MS	NOSQL
Thu	Devops	SPM	MS	DL		NOSQL	SPM	Devops
Fri	DL	SPM	MS	NOSQL	Java Full Stack Technologies Lab			

STAFF DETAILS:

S. No.	Course Code	Course Name	A	B	C	D
1.	V20CSTPE11	Elective – III: Deep Learning	Mr. N.V. Ratnakishor Gade	Mr. N.V. Ratnakishor Gade	Dr. A. Daveedu Raju	Dr. A. Daveedu Raju
2.	V20CSTPE14	Elective – IV: NoSQL Databases	Dr. G Tej Varma	-	-	Dr. G Tej Varma
3.	V20CSTPE16	Elective – IV: Cloud Computing	-	Mr. M.V.V. Krishna	Mr. M.V.V. Krishna	-
4.	V20CSTPE17	Elective-V: Software Project Management	-	-	-	Mr. A. Rajesh
5.	V20CSTPE20	Elective-V: Social Networks and Semantic Web	Mrs. D. Suvarna Lakshmi Manikanteswari	Mrs. D. Suvarna Lakshmi Manikanteswari	Mr. P Rishikesh Sri Tanay	-
6.	V20MBT52	Management Science	Mrs. V Sandhya	Mrs. P Devi	Mrs. K Lalitha Bhavani	Mr. K Pavan Kumar
7.	V20CSTJE03	Job Oriented Elective –III: Full Stack Technologies (FST)	Dr. K. Shirin Bhanu / Mrs. D. Suvarna Lakshmi Manikanteswari		Dr. K. Shirin Bhanu / Mr. P Rishikesh Sri Tanay	
8.	V20CSTJE04	Job Oriented Elective –IV: DevOps	Mr. N.V.Murali Krishna Raja	Mr. N.V.Murali Krishna Raja	Mrs. M.N.V. Surekha	Mrs. M.N.V. Surekha
9.	V20SOC05	Skill Advanced Course - V	Mr. N V M K Raja	Mrs. D. Suvarna Lakshmi Manikanteswari	Mr. M.V.V. Krishna	Dr. A. Daveedu Raju



Head of the Department

COURSE STRUCTURE

VII - SEMESTER

S.N o.	Course Code	Name of the Course	L	T	P	C	
1	Professional Elective-III		PEC	3	0	0	3
	V20CSTPE09	Advanced Computer Architecture					
	V20CSTPE10	Big Data Analytics					
	V20CSTPE11	Deep Learning					
	V20CSTPE12	Human Computer Interaction					
2	Professional Elective-IV		PEC	3	0	0	3
	V20CSTPE13	Design Patterns					
	V20CSTPE14	NoSQL Databases					
	V20CSTPE15	Reinforcement Learning					
	V20CSTPE16	Cloud Computing					
3	Professional Elective-V		PEC	3	0	0	3
	V20CSTPE17	Software Project Management					
	V20CSTPE18	Scripting Languages					
	V20CSTPE19	Natural Language Processing					
	V20CSTPE20	Social Networks and Semantic Web					
4		Open Elective -III / Job Oriented Elective –III	OEC	3	0	0	3
			JOE	0	0	6	
5		Open Elective -IV / Job Oriented Elective – IV	OEC/JOE	3	0	0	3
6	V20MBT52	Management Science	HSS	3	0	0	3
7	V20SOC05	Skill Oriented Course-V*	SOC	1	0	2	2
8	V20CSP02	Mini Project /Internship	Internship	0	0	6	3
Total				16	0	14	23

Total Contact Hours: 30

Total Credits: 23

* The Student need to select one Skill Oriented Course from the given pool of courses.



LESSON PLANS

Deep Learning

Academic Year : 2024-25

Semester : VII

Name of the Course: Deep Learning (Elective-III)

Programme: B.Tech

Sections :A,B,C&D

Course Code: V20CSTPE11

COURSE OUTCOMES (Along with Knowledge Level): After completion of this course, the students will be able to:

S.No.	Course Outcome	BTL
1.	Describe the fundamentals of deep learning.	K2
2.	Illustrate the working of deep feed forward neural networks	K2
3.	Discuss regularization and optimization techniques used in deep neural networks.	K2
4.	Illustrate the working of convolution neural networks.	K2
5.	Explain about recurrent and recursive neural networks.	K2

Text Books:

1. Deep Learning, Ian Goodfellow, YoshuaBengio, and Aaron Courville, MIT Press.

Reference Books:

1. Neural Networks and Deep Learning, Charu C. Aggarwal, Springer.
2. Fundamentals of Deep Learning, Nikhil Buduma, 1st Edition, O'Reilly.

Targeted Proficiency Level (For each course Outcome):

COs		C01	C02	C03	C04	C05
Targeted Proficiency Level		60	60	60	60	60
Targeted level of Attainment	Level 3	60	60	60	60	60
	Level 2	55	55	55	55	55
	Level 1	50	50	50	50	50

Lesson Plan:

S. No	Course Outcome	Intended Learning Outcomes (ILO)	K L of ILO	#Hrs	Pedagogy	Teaching aids
1	CO1	Dissemination of Vision, Mission of the Dept. and PEOs, Pos, & PSOs of the Programme		1	Lecture	ICT
2		Historical Trends in Deep Learning	K2	1	Lecture	ICT
3		The Many Names and Changing Fortunes of Neural Networks	K2	1	Lecture	ICT Chalk & Talk
4		Increasing Dataset Sizes, Model Sizes and Accuracy	K2	2	Lecture with Discussion	ICT Chalk & Talk
5		Complexity and Real-World Impact.	K2	2	Lecture with Discussion	ICT
6	CO2	Deep Feed forward Networks	K2	2	Lecture with Discussion	ICT Chalk & Talk
7		Learning XOR	K2	1	Lecture with Discussion	ICT Chalk & Talk
8		Gradient-Based Learning	K2	2	Lecture with Discussion	ICT
9		Hidden Units, Architecture Design	K2	2	Lecture with Discussion	ICT Chalk & Talk
10		Back Propagation and Other Differentiation Algorithms	K2	2	Lecture with Discussion	ICT Chalk & Talk
11	CO3	Regularization for Deep Learning	K2	2	Lecture with Discussion	ICT Chalk & Talk
12		Parameter Norm Penalties, Early Stopping, Dropout	K2	2	Lecture with Discussion	ICT Chalk & Talk
13		Optimization for Training Deep Models	K2	2	Lecture with Discussion	ICT Chalk & Talk
14		How Learning Differs from Pure Optimization?	K2	2	Lecture with Discussion	ICT
15		Challenges, Basic Algorithms, Parameter Initialization Strategies	K2	3	Lecture with Discussion	ICT
16		Algorithms with Adaptive Learning Rates	K2	3	Lecture	ICT Chalk & Talk
17		Optimization Strategies and Meta-Algorithms	K2	2	Lecture	ICT Chalk & Talk
18	CO4	Convolution Networks: The Convolution Operation, Motivation, Pooling	K2	3	Lecture	ICT Chalk & Talk
19		Convolution and Pooling as an Infinitely StrongPrior,	K2	1	Lecture	ICT Chalk & Talk
20		Variants of Basic Convolution Functions, Structured Outputs	K2	2	Lecture with Discussion	ICT Chalk & Talk
21		Data Types, Efficient Convolution Algorithms, Random or	K2	2	Lecture with Discussion	ICT Chalk & Talk

		Unsupervised Features				
22		The Neuro-scientific Basis for Convolution Networks,	K2	1	Lecture with Discussion	ICT Chalk &Talk
23		Convolution Networks and the History of Deep Learning.	K2	2	Lecture with Discussion	ICT Chalk &Talk
24	CO5	Sequence Modelling- Recurrent and Recursive Nets	K2	2	Lecture	ICT
25		Unfolding Computational Graphs, Recurrent Neural Networks,	K2	2	Lecture with Discussion	ICT Chalk &Talk
26		Bidirectional RNNs, Encoder-Decoder Sequence-to-Sequence Architectures	K2	3	Lecture with Discussion	ICT Chalk &Talk
27		Deep Recurrent Networks, Recursive Neural Networks	K2	3	Lecture with Discussion	ICT Chalk &Talk
28		The Challenge of Long-Term Dependencies, Echo State Networks,	K2	3	Lecture with Discussion	ICT Chalk &Talk
29		Leaky Units and Other Strategies for Multiple Time Scales	K2	1	Lecture with Discussion	ICT
30		LSTM and Other Gated RNNs, Explicit Memory.	K2	2	Lecture with Discussion	ICT Chalk & Talk

Total No. of Classes: 60

NoSQL Databases

Academic Year : 2024-25

Year/ Semester : VII Sem

Name of the Course: NoSQL Databases (Elective –IV)

Programme: B.Tech

Section: A, B, C, D

Course Code: V20CSTPE14

Course Outcomes (Along with Knowledge Level): After completion of this course, the students will be able to:

S.No.	Course Outcome	BTL
1.	Discuss four types of NoSQL Databases (Document-oriented, Key Value Pairs, Column oriented and Graph)	K2
2.	Illustrate Replication and Sharding	K2
3.	Explain NoSQL Key/Value databases using MongoDB	K2
4.	Demonstrate Column – oriented NoSQL databases using Apache HBASE.	K3
5.	Explain Graph NoSQL databases using Neo4.	K3

Text Books:

1. NoSQL Distilled : A Brief Guide to the Emerging World of Polyglot Persistence, 1st Edition, 2012. Authors : Sadalage P. & Fowler, Publication : Pearson Education.
2. The Definitive Guide to MongoDB : A complete guide to dealing with Big Data using MongoDB, 3rd Edition 2015. Authors: Eelco Plegge, David Hows, Peter Membrey, Tim Hawkins, Apress Publishers.

Reference Books:

1. Redmond, E. Wilson, Perkins : Seven Databases in Seven Weeks : A Guide to Modern Databases and the NoSQL Movement Edition : 2nd, 2018, O'Reilly Publishers

Targeted Proficiency and attainment Levels (for each Course Outcome):

Cos		CO1	CO2	CO3	CO4	CO5
Targeted Proficiency Level		60	60	60	60	60
Targeted level of Attainment	Level 3	60	60	60	60	60
	Level 2	50	50	50	50	50
	Level 1	40	40	40	40	40

Lecture Plan:

S#	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1.	CO1	Dissemination of Vision, Mission of the Dept. and PEOs, Pos, &PSOs of the Programme		1	Lecturer	ICT Chalk &Talk
2.		Overview and History of NoSQL Databases Definition of the Four types of NoSQL Database, Values of Relational Database	K2	1	Lecturer	ICT Chalk &Talk
3		Persistent Data, Integration, Impedance Mismatch, Clusters Attack, Emergence of NoSQL	K2	2	Lecturer	ICT Chalk &Talk
4		Comparison of Relational Databases to new NoSQL stores, MongoDB,	K2	2	Lecturer	ICT Chalk &Talk
5		Introduction to Cassandra, HBASE, Neo4j use and deployment	K2	2	Lecturer	ICT Chalk &Talk
6		Challenges NoSQL approach, Explanation about four types of NoSQL databases	K2	3	Lecturer	ICT Chalk &Talk
Total				11		ICT Chalk &Talk
S#	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
7	CO2	Replication and sharding	K2	2	Lecturer	ICT Chalk &Talk
8		Map Reduce on databases	K2	1	Lecturer	ICT Chalk &Talk
9		Distribution Models	K2	2	Lecturer	ICT Chalk &Talk
10		Single Server	K2	1	Lecturer	ICT Chalk &Talk
11		Sharding	K2	1	Lecturer	ICT Chalk &Talk
12		Master-Slave Replication	K2	2	Lecturer	ICT Chalk &Talk
13		Peer-to-Peer Replication	K2	2	Lecturer	ICT Chalk &Talk
14		Combining Sharding andReplication	K2	2	Lecturer	ICT Chalk &Talk
Total				13		

S#	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teachingaids
15	CO 3	NoSQL Key/Value databases using MongoDB	K2	2	Lecturer	ICT Chalk &Talk
16		Document Databases, Features, Use Cases	K2	2	Lecturer	ICT Chalk &Talk
17		Event Logging , Content Management Systems, Blogging Platforms, Web Analytics	K2	3	Lecturer	ICT Chalk &Talk
18		E-Commerce Applications	K3	1	Lecturer	ICT Chalk &Talk
19		Complex Transactions Spanning , Different Operations	K2	3	Lecturer	ICT Chalk &Talk
20		Aggregate Structure and its Queries	K2	3	Lecturer	ICT Chalk &Talk
Total				14		
S#	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
21	CO 4	Column-oriented NoSQL databases using Apache HBASE	K3	2	Lecturer	ICT Chalk &Talk
22		Column-oriented NoSQL databases using Apache Cassandra	K3	2	Lecturer	ICT Chalk &Talk
23		Architecture of HBASE	K3	1	Lecturer	ICT Chalk &Talk
24		Columnly-Family Data store and its features – Consistency, Transactions, Availability, Query Features	K3	2	Lecturer	ICT Chalk &Talk
25		Scaling, Use cases	K3	1	Lecturer	ICT Chalk &Talk
26		Event Logging , Content Management Systems, Blogging Platforms	K3	2	Lecturer	ICT Chalk &Talk
27		Counters, Expiring Usage, When to use	K3	2	Lecturer	ICT Chalk &Talk
Total				12		

S#	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
28	CO 5	Graph NoSQL databases usingNeo-4	K3	2	Lecturer	ICT Chalk &Talk
29		NoSQL database development tools and programminglanguages	K3	2	Lecturer	ICT Chalk &Talk
30		Graph Databases Features – Consistency, Transactions, Availability, Query Features	K3	2	Lecturer	ICT Chalk &Talk
31		Scaling , Use cases	K3	1	Lecturer	ICT Chalk &Talk
32		Connected Data, Routing	K3	1	Lecturer	ICT Chalk &Talk
33		Dispatch and Location-BasedServices	K3	2	Lecturer	ICT Chalk &Talk
34		Recommendation Engines	K3	1	Lecturer	ICT Chalk &Talk
35		When to use	K3	1	Lecturer	ICT Chalk &Talk
Total				12		

Total No. of Classes: 62

Cloud Computing

Academic Year: 2024-25

Year/ Semester: VII

Name of the Course: Cloud Computing

Programme: B.Tech

Section: A,B,C,D

Course Code : V20CSTPE16

Course Outcomes (Along with Knowledge Level):After successful completion of course the student will able to

S.No.	Course Outcome	BTL
1.	Explain the basic concepts of Cloud Computing.	K2
2.	Describe the Virtualization and Migration concepts of Cloud.	K2
3.	Explain the Cloud Application Design Methodologies.	K2
4.	Illustrate the security aspects of Cloud.	K2
5.	Illustrate the SLA management aspects of Cloud.	K2

Text Books:

1. Cloud Computing: Principles and Paradigms, Rajkumar Buyya, James Borberg, Andrzej Goscinski, Wiley Publication.
2. Cloud Computing: A Hands –on Approach, Arshdeep Bahga, Vijay Mediseti, University Press.

Reference Books:

1. Cloud Computing – Web Based Applications That Change the Way you Work and Collaborate Online, Michael Miller, Pearson Education.
2. Cloud Computing: A Practical Approach, Antony T.Velte, Toby J.Velte, Robert Elsenpeter, McGraw-Hill, (2010).

Targeted Proficiency and attainment Levels (for each Course Outcome):

Cos		CO1	CO2	CO3	CO4	CO5
Targeted Proficiency Level		65	60	60	65	60
Targeted level of Attainment	Level 3	60	60	60	60	60
	Level 2	50	50	50	50	50
	Level 1	40	40	40	40	40

Lecture Plan:
UNIT-I

S. No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO1	Introduction to OBE, Dissemination of Vision, Mission of the Dept. and PEOs, POs & PSOs of the Programme.		1	Lecture	BB+ICT
2		Define the Cloud Computing and Explain the Types of clouds and Layers in Clouds	K1	2	Lecture with discussion	BB+ICT
3		Identify the Desired features of a Cloud.	K1	1	Lecture	BB+ICT
4		Describe the Infrastructure as a Service Providers (IaaS).	K1	2	Lecture with discussion	BB+ICT
5		Describe the Platform as a Service Providers (PaaS).	K1	2	Lecture with discussion	BB+ICT
6		Identify the Challenges and Risks in Cloud Computing	K2	1	Lecture	BB+ICT
		Total		9 Hrs		

UNIT-II

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO2	Outline The Concepts and enabling technologies of cloud computing	K1	1	Lecture	BB
2		Explain the Virtualization and its types.	K2	1	Lecture	BB
3		Describe the need for Load Balancing and Outline the Algorithms used.	K2	1	Lecture	BB
4		Define Replication and its types	K2	1	Lecture	BB
5		Define SDN, and SDN Architecture Key elements	K1	1	Lecture	BB
6		Explain NFV in relationship to SDN and NFV Architecture	K2	2	Lecture with Discussion	BB+ICT
7		Demonstrate the seven step model of migration into a cloud	K2	2	Lecture with Discussion	BB+ICT
8		Discuss the Migration mitigation and Risks.	K2	1	Lecture with Discussion and in class Assignment	BB+ICT
		Total		10 Hrs		

UNIT-III

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO3	Outline Verification and Validation activities	K1	1	Lecture	BB
2		Explain the Design Considerations for Cloud Applications.	K2	1	Lecture	BB
3		Explain Reference Architectures for Cloud Applications.	K2	1	Lecture	BB
4		Demonstrate Cloud Application Design Methodologies: SOA	K2	1	Lecture	BB
5		Explain Cloud Component Model.	K2	2	Lecture with Discussion	BB+ICT
6		Demonstrate MVC	K2	1	Lecture with Discussion and in class Assignment	BB+ICT
7		Illustrate Data Storage Approaches.	K2	2	Lecture with Discussion	BB+ICT
		Total		9 Hrs		

UNIT-IV

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO4	Outline Cloud Security	K1	1	Lecture	BB
2		Explain the Cloud Security Architecture (CSA).	K2	2	Lecture	BB
3		Interpret Authentication, Authorization, and Identity.	K2	3	Lecture	BB
4		Explain Access Management.	K2	2	Lecture	BB
5		Demonstrate Data Security, Key Management	K2	2	Lecture with Discussion and in class Assignment	BB+ICT
		Total		10 Hrs		

UNIT-V

S. No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO5	Outline SLA Management in Cloud Computing	K1	1	Lecture	BB
2		Explain the Service Level Agreements (SLA).	K2	2	Lecture	BB
3		Interpret Traditional Approaches to SLO Management.	K2	2	Lecture	BB
4		Explain Types of SLA.	K2	2	Lecture	BB
5		Discuss Life Cycle of SLA	K2	1	Lecture with Discussion and in class Assignment	BB+ICT
6		Demonstrate SLA Management in Cloud	K2	2	Lecture with Discussion	BB+ICT
		Total		10 Hrs		

Total No. Of Classes: 48

Software Project Management

Academic Year: 2024-25

Year/ Semester: VII

Name of the Course: Software Project Management
:V20CSTPE17

Programme: B.Tech

Section: A,B,C,D

Course

Code

Course Outcomes (Along with Knowledge Level): After successful completion of course the student will be able to

S.No.	Course Outcome	Knowledge Level
1.	Describe Software Project Management Terminology.	K2
2.	Explain various Software development process models and Software Lifecycle phases.	K2
3.	Illustrate various Effort Estimation Techniques and activity network models for Project Planning.	K3
4.	Demonstrate Risk Management Concepts and resource allocation.	K3
5.	Explain importance of project monitoring and control for accomplishing project goals and software quality.	K2

Text Books:

1. Software Project Management, Bob Hughes & Mike Cotterell, 6th Ed, TMH.
2. Software Project Management, Walker Royce, Pearson Education, 2005.

Reference Books:

1. Software Project Management in Practice, Pankaj Jalote, 9th Ed Pearson.
2. Software Project Management, Joel Henry, 3rd Ed, Pearson Education.

Targeted Proficiency and attainment Levels (for each Course Outcome):

Cos		CO1	CO2	CO3	CO4	CO5
Targeted Proficiency Level		60	60	60	60	60
Targeted level of Attainment	Level 3	70	70	70	70	70
	Level 2	65	65	65	65	65
	Level 1	60	60	60	60	60

Lecture Plan

UNIT-I

S.No.	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1		Dissemination of Vision, Mission of the Dept and PEOs, Pos,& PSOs of the Programme			Lecture	BB+ICT
2	CO1	Differentiate Software Project and other type of projects	K2	1	Lecture	BB+ICT
3		Describe software project management activities.	K1	1	Lecture	BB+ICT
4		Discuss various Categories in software Projects	K2	1	Lecture	BB+ICT
5		Identify types of stake holders, objectives and goals in software project management.	K1	2	Lecture	BB+ICT
6		Describe Stepwise project planning, project scope , Objectives and infrastructure.	K2	2	Lecture	BB+ICT
7		Identify Project products, Deliverables, activities and effort estimation.	K1	1	Lecture	BB+ICT
		Total		8		

UNIT-II

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO2	Describe Build or buy approach	K1	1	Lecture	BB
2		Describe Process Models: Waterfall , Prototyping and Incremental	K2	2	Lecture	BB+ICT
3		Discuss Agile methods: Extreme programming, Atern method	K2	1	Lecture	BB+ICT
4		Select an appropriate model	K2	1	Lecture	BB
5		Classify Project Life Cycle Phases.	K2	2	Lecture	BB+ICT
		Total		7		

UNIT-III

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO3	Describe Software Effort Estimation Techniques.	K1	1	lecture	BB
2		Discuss Function Point Analysis.	K2	1	lecture with Discussion	BB +ICT
3		Explain SLOC: Software Metrics and Measurements.	K2	1	lecture	BB + ICT
4		Demonstrate COCOMO: A Parametric Model	K3	2	lecture	BB + ICT
5		Demonstrate Use-Case based Estimation Techniques.	K3	1	lecture with Discussion	BB +ICT
6		Explain various Activity Identification Approaches: Sequencing and Scheduling Activities.	K2	1	lecture	BB +ICT
7		Illustrate Network Planning Models in Project Scheduling: Forward pass and Backward pass and Critical Path Analysis.	K3	2	lecture with Discussion	BB +ICT
		Total		9		

UNIT-IV

S. No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO4	Describe various Risk Management Categories.	K1	2	Lecture	BB
2		Demonstrate concepts of Risk Identification, Assessment, Planning and Management.	K3	2	Lecture with discussion	BB+ICT
3		Demonstrate PERT Technique.	K3	1	Lecture	BB
5		Describe Resource Allocation types, Resource requirement and resource scheduling	K2	2	Lecture	BB
		Total		7		

UNIT-V

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO5	Describe the concept of Project Monitoring and Control.	K1	1	lecture	BB
2		Describe Data collection, Visualizing progress	K1	1	lecture	BB
3		Explain Cost monitoring and Earned Value Analysis.	K2	2	lecture	BB + ICT
4		Define Software Quality.	K1	1	Lecture	BB + ICT
5		Describe importance of quality and ISO 9126.	K1	1	lecture	BB
6		Explain the concepts of product Quality and Process Quality.	K2	1	lecture with discussion	BB
7		Describe Statistical Process Control Capability Maturity Model.	K2	1	Lecture with Discussion	BB
8		Discuss various Techniques to Enhance Software Quality.	K2	1	Lecture with Discussion	
		Total		9		

Total No. of Classes: 60

Social Network and Semantic Web

Academic Year: 2024-25

Year/ Semester: VII

Name of the Course: Social Network and Semantic Web

Programme: B.Tech

Section: A,B,C,D

Course Code : V20CSTPE20

Course Outcomes (Along with Knowledge Level): After successful completion of course the student will be able to

S.No.	Course Outcome	BTL
1.	Demonstrate knowledge by explaining the three different named generations of web	K3
2.	Construct a Social Network	K3
3.	Relate Knowledge representation methods for semantic web [K3]	K3
4.	Describe web services and its Applications	K2
5.	Develop “Linked data” Application using semantic web Technologies	K3

Text Books:

1. Social Networks and the Semantic Web, Peter Mika, Springer, 2007.
2. Semantic Web Technologies, Trends and Research in Ontology based systems, J. Davies, Rudi Studer, Paul Warren, John Wiley & Sons.

Reference Books:

1. Semantic Web and Semantic Web Services – Liyang Lu Chapman and Hall/CRC Publishers, (Taylor & Francis Group)
2. Information Sharing on the semantic Web – Heiner Stuckenschmidt; Frank Van Harmelen, Springer Publications

Targeted Proficiency and attainment Levels (for each Course Outcome):

Cos		CO1	CO2	CO3	CO4	CO5
Targeted Proficiency Level		60	60	60	60	60
Targeted level of Attainment	Level 3	70	70	70	70	70
	Level 2	65	65	65	65	65
	Level 1	60	60	60	60	60

Lecture Plan

UNIT-I

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1		Dissemination of Vision, Mission of the Dept and PEOs, Pos,& PSOs of the Programme			Lecture	BB
2	CO1	Explain The Semantic web	K2	1	Lecture	BB
3		Describe Limitations of the current Web	K1	2	Lecture	BB
4		Describe The Semantic Solution	K1	2	Lecture	BB
5		Explain Development of the semantic web	K2	2	Lecture	BB
6		Explain Emergence of social web	K2	1	Lecture	BB
		Total		8		

UNIT-II

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO2	Describe What is network analysis? Development of Social Network Analysis	K1	1	Lecture	BB
2		Explain Key concepts and measures in network analysis	K2	2	Lecture	BB
3		Explain Electronic sources for network analysis	K2	2	Lecture	BB
4		Discuss Electronic discussion networks	K2	1	Lecture	BB
5		Discuss Blogs and online communities	K2	2	Lecture	BB+ICT
6		Explain Web-based networks	K2	2	Lecture	BB
		Total		10		

UNIT-III

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO3	Describe Knowledge Representation on the Semantic Web	K1	1	lecture	BB
2		Explain Ontologies	K2	2	lecture with Discussion	BB
3		Discuss Ontologies and their role in the Semantic Web	K2	2	lecture	BB + ICT
4		Discuss Ontology languages for the semantic Web	K2	2	lecture	BB + ICT
5		Explain Ontology languages for the semantic Web	K2	2	lecture with Discussion	BB
6		Modeling and Aggregating Social Network Data: Describe Modeling and Aggregating Social Network Data	K1	2	Lecture	BB
7		Classify State of the art in network data representation	K2	2	Lecture with discussion	BB+ICT
8		Explain Ontological representation of Social individuals	K2	2	Lecture	BB
9		Discuss Ontological representation of social relationships	K2	1	Lecture	BB
10		Explain Aggregating and reasoning with social network data.	K2	2	Lecture	BB
		Total		18		

UNIT-IV

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO4	Identify developing social semantic applications	K1	1	lecture	BB
2		Discuss Building Semantic Web applications with social network features	K2	2	lecture	BB
3		Outline Flink- the social networks of the Semantic Web community	K1	2	lecture	BB + ICT
4		Explain Open academia	K2	1	Lecture	BB + ICT
5		Explain Open academia distributed	K2	1	lecture	BB
6		Discuss semantic-based publication management.	K2	2	lecture with discussion	BB
7		Discuss semantic-based publication management.	K2	1	Lecture with practical	BB
		Total		18		

UNIT-V

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO5	Demonstrate the Evaluation of Web-Based Social Network Extraction	K3	1	Lecture	BB + ICT
2		Outline Differences between survey methods and electronic data extraction	K1	1	Lecture	BB + ICT
3		Discuss context of the empirical study	K1	2	Lecture	BB + ICT
4		Describe Data collection	K1	2	Lecture with Discussion	BB + ICT
5		Describe Preparing the data	K1	1	Lecture	BB + ICT
6		Explain optimizing goodness of fit	K2	1	Lecture	BB + ICT
7		Explain Comparison across methods and network	K2	2	Lecture with Discussion	BB + ICT
8		Discuss Predicting the goodness of fi	K2	2	Lecture with Discussion	BB + ICT
9		Discuss Evaluation through analysis	K2	2	Lecture with Discussion	BB + ICT
		Total		14		

Total No. of Classes: 60

Management Science

Academic Year: 2024-25

Programme: B.Tech

Year/ Semester: VII

Section: A,B,C,D

Name of the Course: Management Science

Course Code : V20MBT52

Course Outcomes (Along with Knowledge Level): After successful completion of course the student will be able to

S.No.	CO No.	Course Outcome	BTL
1.	CO1	Understand various approaches to Management.	K2
2.	CO2	To get familiarity with operations management in an organization	K2
3.	CO3	Understand the Functions of Human Resource Management, Marketing Management and Financial Management	K2
4.	CO4	To Sketch the networks for better project management	K3
5.	CO5	Understand the Concept of Strategic Management and to get familiarity with contemporary developments in business management.	K2

Text Books:

1. Dr. P. Vijayakumar & Dr. N. Apparao, Management Science, cengage,
2. Dr. AR. Arysri, Management Science, TMH2011

Reference Books:

1. Philip Kotler & Armstrong: Principles of Management, Pearson publications.
2. Hitt and vijayakumar: strategic Management, cengage learning

Targeted Proficiency and attainment Levels (for each Course Outcome):

Cos		CO1	CO2	CO3	CO4	CO5
Targeted Proficiency Level		60	60	60	60	60
Targeted level of Attainment	Level 3	60	60	60	60	60
	Level 2	55	55	55	55	55
	Level 1	50	50	50	50	50

Lecture Plan:

S. No.	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours Required	Pedagogy	Teaching aids
1	CO1	Definitions of management,	K1	1	Lecture	BB
2		Describe the Functions of management.	K1	2	Lecture	BB
3		Evaluation of management thought	K2	2	Lecture+ discussion	BB
4		Explain Theories of motivation.	K2	2	Lecture	BB
5		Managerial skills	K2	1	Lecture + discussion	BB
6		Types of organization structures	K2	1	Lecture + discussion	BB
7		State the International Management structure.	K2	2	Lecture + discussion	BB
1	CO2	Plant Location and Layout	K1	1	Lecture	BB
2		Work Study and Statistical Quality Control.	K2	2	Lecture + discussion	BB
3		Control charts (P- chart, R chart, and C chart)	K3	2	Lecture + discussion	BB
4		Explain the concept of Material Management	K2	1	Lecture + discussion	BB
5		Need for inventory control	K2	2	Lecture + discussion	BB
6		EOQ, ABC analysis simple problems and Types of ABC analysis.	K3	3	Lecture + discussion	BB
1	CO 3	Concept of HRM, HRD and PMIR	K1	1	Lecture	BB
2		Functions of HR manager	K2	2	Lecture + discussion	BB
3		Job Analysis	K2	1	Lecture + discussion	BB
4		Job Evaluation and Merit Rating methods.	K2	2	Lecture + discussion and In-class Assignment	BB+PPT
5		Marketing Management ,Functions of Marketing	K2	1	Lecture + discussion	BB
6		Four P's of marketing ,New product Development	K2	1	Lecture + discussion	BB
7		Product life cycle, Service Marketing	K2	2	Lecture + discussion and In-class Assignment	BB+PPT
1	CO 4	Construction of Network	K2	3	Lecture+ discussion	BB
2		Difference between PERT and CPM and Problems	K2	4	Lecture + discussion and In-class Assignment	BB
3		Compute Critical path, Probability,	K3	4	Lecture +	BB

		Project crashing (Simple problems)			discussion	
1	CO5	Describe Vision, Mission, Goals and Strategy.	K2	2	Lecture + discussion	BB
2		Describe Strategic Management Process	K2	2	Lecture + discussion	BB
3		Discuss Corporate Planning	K2	2	Lecture + discussion and In-class Assignment	BB+PPT
4		Explain Environmental Scanning	K2	1	Lecture	BB
5		Describe SWOT analysis.	K2	1	Lecture + discussion and In-class Assignment	BB+PPT
6		Describe the Concept of ERP	K2	1	Lecture + discussion and In-class Assignment	BB+PPT
7		Describe the concept of Total Quality Management	K1	1	Lecture	BB
8		Describe the concept of Six sigma	K2	1	Lecture + discussion	BB
9		Describe the concept of Supply chain Management.	K2	1	Lecture	BB
10		Describe the concept of Business process out sourcing	K2	1	Lecture + discussion	BB
11		Explain Lean Start-ups and Entrepreneurship	K2	1	Lecture + discussion and In-class Assignment	BB+PPT

Total No. Of Classes: 57

Full Stack Technologies

Academic Year: 2024-25

Year/ Semester: VII

Name of the Course: Full Stack Technologies

Programme: B.Tech

Section: A,B,C,D

Course Code : V20CSTJE03

Course Outcomes (Along with Knowledge Level): After successful completion of course the student will be able to

CO	Course Outcomes	Knowledge Level
CO1	Demonstrate IDE tools Installation	K3
CO2	Develop programs using servlets.	K3
CO3	Illustrate MVC architecture.	K3
CO4	Demonstrate applications of Hibernate.	K3
CO5	Illustrate Spring MVC Framework.	K3

Targeted Proficiency and Attainment Levels (for each course Outcome):

Cos		CO1	CO2	CO3	CO4	CO5
Targeted Proficiency Level		70	70	70	70	70
Targeted level of Attainment	Level 3	70	70	70	70	70
	Level 2	60	60	60	60	60
	Level 1	50	50	50	50	50

Lecture Plan:

S. No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO 1	Dissemination of COs	-	1	Lecture With Discussion	ICT
2		Basic Installation of IDEs and Development Tools	K3	2	Lecture With Discussion	ICT ICT ICT
3						
		Total		3		

S. No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO 2	• Introduction to Servlets. • Develop Servlet application to print current date & time. • Develop Servlet program to link Html & Servlet Communication. • Develop Servlet program to Auto refresh a page. • Demonstrate session tracking u • Develop Servlet program to insert/delete/update the record into database. • Develop Servlet program to add cookie to selected value	K3	9	Lecture With Discussion	ICT
2						
		Total		9		

S. No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO 3	• Introduction to MVC in java. • Develop sample program on Model Layer in MVC Using Java. • Develop sample program on View Layer in MVC Using Java. • Develop sample program on Controller Layer in MVC Using Java. • Demonstrate MVC deployment in java. • Rules for MVC Mapping in Server Side. • How to use Web Server for MVC Deployment.	K3	9	Lecture with Discussion	ICT
2						
		Total		9		

S. No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO 4	• Introduction to Spring MVC. • Demonstrate the usage of Dispatcher Servlet in Spring MVC. • Load the spring jar files or add dependencies in the case of Maven. • Create the controller class. • Provide the entry of controller in the web.xml file. • Define the bean in the separate XML file. • Display the message in the JSP page. • Start the server and deploy the project. • Execute the application on webserver using Spring MVC.	K3	9	Lecture with Discussion	ICT
2						
		Total		9		

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO 5	• Introduction to Hibernate. • What is ORM? Demonstrate the components of Hibernate • How to persist objects using Hibernate and how to use map using XML and Annotations • How to implement Inheritance in Hibernate • Working with relationship between entities-association • Transactions in Hibernate • Querying with HQL (Hibernate Query Language) • Various other forms of querying - Criteria, QBE etc. • Understanding Some Debugging Tools in Java NetBeans, Eclipse, IntelliJ IDEA, Visual Studio Code.	K3	12	Lecture with Discussion	ICT
		Total		12		

Total No. of Classes: 42

DevOps

Academic Year: 2024-25

Year/ Semester: VII

Name of the Course: DevOps

Programme: B.Tech

Section: A,B,C,D

Course Code : V20CSTJE04

Course Outcomes (Along with Knowledge Level): After successful completion of course the student will able to

S.No.	Course Outcome	BTL
1.	Discuss the traditional software development.	K2
2.	Discuss the concepts of rise of agile methodologies.	K2
3.	Discuss the concept of DevOps and Agile.	K2
4.	Demonstrate the purpose of DevOps.	K3
5.	Illustrate the Operations of CAMS.	K2

Text Books:

1. The DevOps Handbook - Book by Gene Kim, Jez Humble, Patrick Debois, and Willis.

Reference Books:

1. What is DevOps? - by Mike Loukides.

Targeted Proficiency and attainment Levels (for each Course Outcome):

Cos		CO1	CO2	CO3	CO4	CO5
Targeted Proficiency Level		65	65	60	60	60
Targeted level of Attainment	Level 3	60	60	60	60	60
	Level 2	50	50	50	50	50
	Level 1	40	40	40	40	40

Lecture Plan:**Unit-I**

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO1	Introduction to OBE, Dissemination of Vision, Mission of the Dept. and PEOs,POs & PSOs of the Programme.		1	Lecture	BB
2		Describe the software crisis problem.	K1	1	Lecture	BB
3		Identify the historical aspects of software development.	K1	1	Lecture	BB
4		Discuss the importance and challenges of software engineering.	K2	2	Lecture with discussion	BB
5		Explain the phases in software development life cycle.	K2	1	Lecture	BB+ICT
6		Describe the characteristics of good software.	K1	1	Lecture	BB+ICT
7		Explain in detail waterfall model.	K2	2	Lecture with Discussion and in class Assignment	BB+ICT
8		Explain the conflicts between developers and IT operations.	K2	1	Lecture	BB+ICT
		Total		10		

Unit- II

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO2	Outline the growth of agile methodologies.	K1	1	Lecture	BB
2		Describe the principles of agile methodology.	K1	1	Lecture	BB
3		Explain the benefits of agile methodology.	K2	1	Lecture	BB
4		Discuss extreme programming agile model	K2	1	Lecture	BB+ICT
5		Explain scrum and DSDM agile models.	K2	1	Lecture	BB+ICT
6		Explain the pros and cons of agile methodologies over waterfall model.	K2	2	Lecture with discussion and in class Assignment	BB+ICT
7		Discuss iterative agile development.	K2	2	Lecture	BB+ICT
8		Explain the agile core values of Individual and team interactions and delivering working software.	K2	1	Lecture	BB
9		Describe the importance of Customer collaboration and handling change request in agile.	K1	1	Lecture	BB
		Total		11		

Unit-III

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO3	Define Devops and their purpose.	K1	2	Lecture	BB
2		Describe the life cycle of Devops.	K2	2	Lecture with Discussion and in class Assignment	BB+ICT
3		Explain the benefits of Devops.	K2	1	Lecture	BB+ICT
4		Describe the key components of Devops culture.	K2	2	Lecture	BB
5		Explain the similarities between Devops and agile.	K2	2	Lecture	BB+ICT
6		Explain the differences between Devops and agile.	K1	2	Lecture with Discussion and in class Assignment	BB+ICT
		Total		11		

Unit- IV

S. No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO4	Define Minimum Viable Product.	K1	1	Lecture	BB
2		Explain the process to build MVP.	K2	1	Lecture	BB
3		Explain the advantages and drawbacks of MVP.	K2	2	Lecture	BB
4		Differentiate MVP and Prototype.	K2	2	Lecture	BB+ICT
5		Demonstrate the process of Continuous Integration	K3	2	Lecture	BB+ICT
6		Explain the benefits of Continuous Integration	K2	1	Lecture	BB+ICT
7		Demonstrate the role of Devops for CI.	K3	2	Lecture with Discussion and in class Assignment	BB+ICT
8		Define the need of continuous delivery	K1	1	Lecture	BB+ICT
9		Differentiate continuous delivery and continuous deployment.	K2	2	Lecture	BB+ICT
		Total		14		

Unit-V

S.No	Course Outcome	Intended Learning Outcomes (ILO)	Knowledge Level of ILO	No. of Hours	Pedagogy	Teaching aids
1	CO5	Explain the core values (CAMS) of Devops.	K2	2	Lecture	BB+ICT
2		Define Test driven development and its benefits.	K1	1	Lecture	BB
3		Explain Test driven development process.	K2	1	Lecture	BB
4		Differentiate traditional configuration management and Devops configuration management .	K2	1	Lecture	BB+ICT
5		Describe the benefits of Devops configuration management.	K1	1	Lecture	BB
6		Discuss the challenges in infrastructure automation in Devops.	K2	1	Lecture with Discussion and in class Assignment	BB+ICT
7		Describe the benefits of infrastructure as a code in Devops.	K1	1	Lecture	BB
8		Define root cause analysis.	K1	1	Lecture	BB
9		Explain how to perform root cause analysis.	K2	1	Lecture	BB+ICT
10		Explain how to run blamelessness post-mortem.	K2	1	Lecture	BB+ICT
11		Discuss the importance of organizational learning.	K2	1	Lecture	BB+ICT
		Total		12		

Total No. of Classes: 58